

ANALOGVERSE SIMULATION INSTRUCTIONS

Setup Instructions: Participants must utilize Arbitrary Behavioral Voltage Sources (**bv** component in LTspice) to generate complex, multi-frequency test vectors. Two **bv** sources must be placed: one connected to the non-inverting input node (**IN+**) and one to the inverting input node (**IN-**). The equations provided below must be entered into the **V=** field of the respective sources.

Unless otherwise specified,

the simulation directive is a 2-second transient analysis: `.tran 0 2 0 1m`

Phase 1: Ideal Bandpass Validation

These tests isolate specific frequencies to verify the core mathematical transfer function of the circuit.

Test 1: Ideal Alpha Wave Extraction (10 Hz Baseline)

- $V(IN+) = 50u * \sin(2\pi * 10 * time)$
- $V(IN-) = 0$
- **Success Criteria:** A clean, undistorted sine wave output amplified by the designated system gain and centered precisely at the designed DC bias voltage (e.g., 1.65V for a 3.3V ADC).

Test 2: Low-Frequency Out-of-Band Rejection (2 Hz)

- $V(IN+) = 50u * \sin(2\pi * 2 * time)$
- $V(IN-) = 0$
- **Success Criteria:** The output amplitude must be severely attenuated compared to the Test 1 baseline.

Test 3: High-Frequency Out-of-Band Rejection (30 Hz)

- $V(IN+) = 50u * \sin(2\pi * 30 * time)$
- $V(IN-) = 0$
- **Success Criteria:** The output amplitude must be severely attenuated compared to the Test 1 baseline, confirming low-pass roll-off.

Test 4: Differential 50 Hz Notch Verification

- $V(IN+) = 500m * \sin(2\pi * 50 * time)$
- $V(IN-) = 0$
- **Success Criteria:** The output must remain near the DC bias baseline. The 50 Hz signal must be heavily attenuated, confirming notch filter placement.

Phase 2: Instrumentation & Artifact Stress Tests

These tests introduce physiological and environmental interference to validate real-world operational stability.

Test 5: DC Saturation and Blocking Test

- $V(IN+) = 300m + 50u * \sin(2\pi * 10 * time)$
- $V(IN-) = 0$
- **Success Criteria:** The circuit must successfully strip the 300 mV DC offset without rails clipping. The output must identical to Test 1.

Test 6: Pure Common-Mode Rejection Ratio (CMRR) Test

- $V(IN+) = 500m * \sin(2\pi * 50 * time)$
- $V(IN-) = 500m * \sin(2\pi * 50 * time)$
- **Success Criteria:** The output must be a flat line at the DC bias level. Any visible 50 Hz oscillation indicates a failure in the input stage CMRR.

Test 7: Common-Mode + DC Offset + Signal Recovery

- $V(IN+) = 100m + 500m * \sin(2\pi * 50 * time) + 50u * \sin(2\pi * 10 * time)$
- $V(IN-) = 100m + 500m * \sin(2\pi * 50 * time)$
- **Success Criteria:** The circuit must reject the shared 100 mV DC offset and the shared 500 mV 50 Hz mains hum, successfully extracting and amplifying only the 10 Hz signal.

Test 8: Low-Frequency Motion Artifact Interference (1 Hz)

- $V(IN+) = 1m * \sin(2\pi * 1 * time) + 50u * \sin(2\pi * 10 * time)$
- $V(IN-) = 0$
- **Success Criteria:** The massive 1 mV, 1 Hz rolling wave must be flattened by the high-pass filter. The output should not show baseline wander.

Test 9: High-Frequency EMG Muscle Noise Interference (100 Hz)

- $V(IN+) = 100u * \sin(2\pi * 100 * time) + 50u * \sin(2\pi * 10 * time)$
- $V(IN-) = 0$
- **Success Criteria:** The 100 Hz high-frequency fuzz must be smoothed out by the low-pass filter, leaving a clean 10 Hz trace.

Test 10: The Comprehensive Real-World Transient Test

- $V(IN+) = 200m + 500m * \sin(2\pi * 50 * time) + 1m * \sin(2\pi * 1 * time) + 100u * \sin(2\pi * 100 * time) + 50u * \sin(2\pi * 10 * time)$
- $V(IN-) = 200m + 500m * \sin(2\pi * 50 * time)$

- **Success Criteria:** Despite 200 mV DC offset, massive 50 Hz common-mode interference, 1 Hz motion wander, and 100 Hz muscle noise, the circuit must isolate and amplify the 10 Hz Alpha wave without power rail saturation.

Phase 3: Frequency Domain Validation

Participants must configure a standard AC voltage source at the input with an AC amplitude of 1 for this test.

Test 11: AC Sweep (Bode Plot)

- **Directive:** `.ac dec 100 0.1 1000`
- **Success Criteria:** The magnitude plot must display a distinct bandpass plateau spanning strictly from 8 Hz to 12 Hz. A sharp, deep notch must be visible exactly at 50 Hz, achieving a minimum of -40 dB attenuation relative to the passband peak.